

Gender direction	Male-biased	Female-biased
0	his, he, His, himself	herself, she, She, her
1	himself, him, Gavin, His	Hatha, midwifery, Midwifery, feminist
2	Mark, Jon, Darren, Luke	Actress, Zumba, Diana, woman
3	Gordon, he, wind, charge	hers, recipe, Challenge, cookbooks
4	1935, 1955, namely, 1958	Roots, Issue, FHM, yoga
5	M, Mickey, KS, Bethesda	Vietnam, Subject, Elle, Ecuador
6	Keys, correct, address, fuel	leap, Embedded, textile, femininity
7	Papers, Categories, wherein, Newark	Botox, LASIK, periodontal, UnityPoint
8	binding, closely, MT, command	Aventura, brunette, HTML, Disclosure
9	82, 92, 91, 86	ASP.NET, committer, Twilight, Seth
10	t, Cisco, Philips, Sharp	preschool, caregivers, homeowners, Preschool
11	Toulouse, Aviv, scored, commended	intersectional, Equality, equality, ASME
12	addressing, segment, inequalities, segments	Wire, loose, anything, Vincents
13	comparison, Hart, 480, refereed	Matthew, independence, couples, LGBTQ
14	manufacturer, organizers, scope, specifications	homeschooling, ligament, loyalty, graduating

Table 5: words closest to top 15 INLP gender directions (FastText representation, biographies dataset).

20,000 most common vocabulary items, and calculate the closest words to w_i (to get male-biased words) as well as the closest words to $-w_i$ (to get female-biased words). The result are presented in Table 5.

The first gender direction seems to capture pronouns. Other gender directions capture socially biased terms, such as “preschool” (direction 10), “cookbooks” (direction 3) or other gender-related terms, such as “LGBTQ” (direction 15) or “femininity” (direction 6). Interestingly, those are mostly female-biased terms. As for the male-biased words, some directions capture surnames, such as “Gordon” and “Aviv”. Other words which were found to be male-biased are less interpretable, such as words specifying years (direction 4), organizational terms such as “Organizers”, “specifications” (direction 14), or the words “Papers”, “Categories” (direction 7). It is not clear if those are the result of spurious correlations/noise, or whether they reflect actual subtle differences in the way the biographies of men and women are written.

Gender rowspace The above analysis focuses on what information do individual gender directions convey. Next, we aim to demonstrate the influence of the final INLP projection on the representation of words. To this end, we rely on the *rowspace* of the INLP matrix P . Recall that the rowspace is the orthogonal complement of the nullspace. As the INLP matrix P projects to the intersection of nullspaces of the gender directions, the complement $P_R := I - P$ projects to the union of rowspaces of individual gender directions. This is a subspace which is spanned by all gender direc-

tions, and thus can be thought of as an empirical gender subspace within the representation space.

For a given word vector w , the “gender norm” – the norm of its projection on the rowspace, $\|P_R w\|$ – is a scalar quantity which can serve as a measure for the gender-bias of the word. We sort the vocabulary by the ratio between the gender norm and the original norm, $\frac{\|P_R w\|}{\|w\|}$ and present the 200 most gendered words (Table 6).

As before, we see a combination of inherently-gendered words (“motherhood”, “women”, “gender”, “masculinities”), socially-biased terms (“teacher”, “raising”, “semiconductors”, “B.Tech”, “IEEE”, “STEM”, “fashion”) and other words whose connection to gender is less interpretable, and potentially represent spurious correlations (“trauma”, “Vitae”, “smile”, “920”, “forgiveness”).

Top words by component on the gender subspace
motherhood, SSHRC, microfinance, preschool, genocide, IFP, CSE, intersectional, student, homeschooling, photoshoot, intersectionality, 920, breastfeeding, STEM, photojournalistic, haiku, kindergarten, FreeOnes, UNESCO, menstrual, turbulence, NTR, ASME, HFN, ECE, IEEE, feminism, noir, Jadavpur, Motherhood, reportage, Contra, TU, WebSphere, counsellor, photovoltaic, J2EE, contraception, university, PEN, masculinities, parenting, EAP, Politecnico, Feminism, trauma, Universiti, counselling, curriculum, Kanpur, women, edits, Pune, Nanjing, ethnographic, Pinterest, surrealist, taught, Hindustan, students, CNRS, Bangalore, Mumbai, consortium, tooth, Vitae, Kindergarten, nanoscale, school, ACL, scholarships, cloud, Goa, NIJC, Montessori, JSPS, scholarship, Neha, DAAD, endometriosis, carrier, UCI, activism, Ambedkar, EECS, semiconductor, scholar, microfluidic, bikini, Raising, teacher, Feminist, vinyasa, NBER, ethnography, Twilight, Sunil, Shankar, viral, earthquake, semiconductors, historiography, vampire, HMO, PSU, bioenergy, historian, Ravi, Breastfeeding, Raman, resettlement, Shweta, ICTs, UNDP, NVIDIA, HIV, Counselling, HEC, KDD, Hyderabad, contraceptive, macro, Ghaziabad, sexuality, CAS, documentary, mic, biography, postdoc, transnationalism, AMD, CFD, B.Tech, physicist, LGBT, parenthood, HKU, HIP, internationalization, M.Tech, BDS, acne, theorist, HPV, Meerut, ageing, smile, Rajesh, psychoeducational, PUNE, grief, AHA, Essays, discourses, secrets, Swati, EPFL, coaching, IIE, Manoj, BIDMC, infertility, fashion, Chicana, Vaishali, Graduation, sociologist, Gender, EA, MIT, teach, gift, IETF, NPPA, counselor, JPL, gender, menopause, LGBTQ, Waseda, perceptions, praxis, birthday, Jawaharlal, fertility, gendered, coverage, stills, PIH, Balaji, Tagged, baking, USM, postpartum, Goenka, Pooja, forgiveness

Table 6: Words by gender norm.